

STOR 155.010 – Midterm 1 Version 1 Solutions

Aniruddhan (Ani) Ganesaraman

September 24, 2026

1 Multiple Choice

1. You come across a *tank bromeliad*, a rainforest plant whose conical leaves collect reservoirs of rain-water. Each of these reservoirs contains a thriving aquatic ecosystem with many species, including frogs, insects, crustaceans, and microbes. You are interested in taking a sample of the organisms living among the plant's many leaves. Which of these is the best example of **cluster sampling**?
 - (A) You expect each leaf's ecosystem to have high variability, while resembling the ecosystems in all the other leaves, so you collect all of the organisms from the first five leaves you see.
 - (B) **You assign numbers to all the leaves, which you think of as “clusters,” choose five of those numbers at random using a simple random sample, and then collect all the organisms from the leaves corresponding to those chosen numbers.**
 - (C) You list all the possible species (mosquitoes, tree frogs, salamanders, etc.). Then you choose a simple random sample, called a “cluster,” from each species to ensure that no species is over- or under-represented.
 - (D) You collect all of the organisms contained in all the leaves of the plant. This is called a census, or equivalently, a cluster.
 - (E) You collect a single random sample from only the microbes. Including too many larger organisms in the sample can lead to “clustering,” which is a form of bias.
2. Researchers want to determine whether listening to music while studying affects performance on a memory test. Because they believe sleep may affect memory, participants are first separated into groups according to whether they slept fewer than 7 hours or at least 7 hours the previous night. Within each group, participants are randomly assigned either to study with music or to study in silence. In this experiment, amount of sleep is:
 - (A) the explanatory variable
 - (B) the response variable
 - (C) **a blocking variable**
 - (D) a treatment
 - (E) a placebo

3. A scientist is investigating whether a new fertilizer affects crop yield compared with an existing fertilizer. She randomly assigns half of the plots on a farm to receive the new fertilizer and the remaining half to receive the existing fertilizer. After one year, she compares the mean crop yields of the two groups. Which statistical term below would you use to describe this scenario?
- (A) A sample survey
 - (B) An observational study
 - (C) **An experiment**
 - (D) A population census
 - (E) Placebo administration

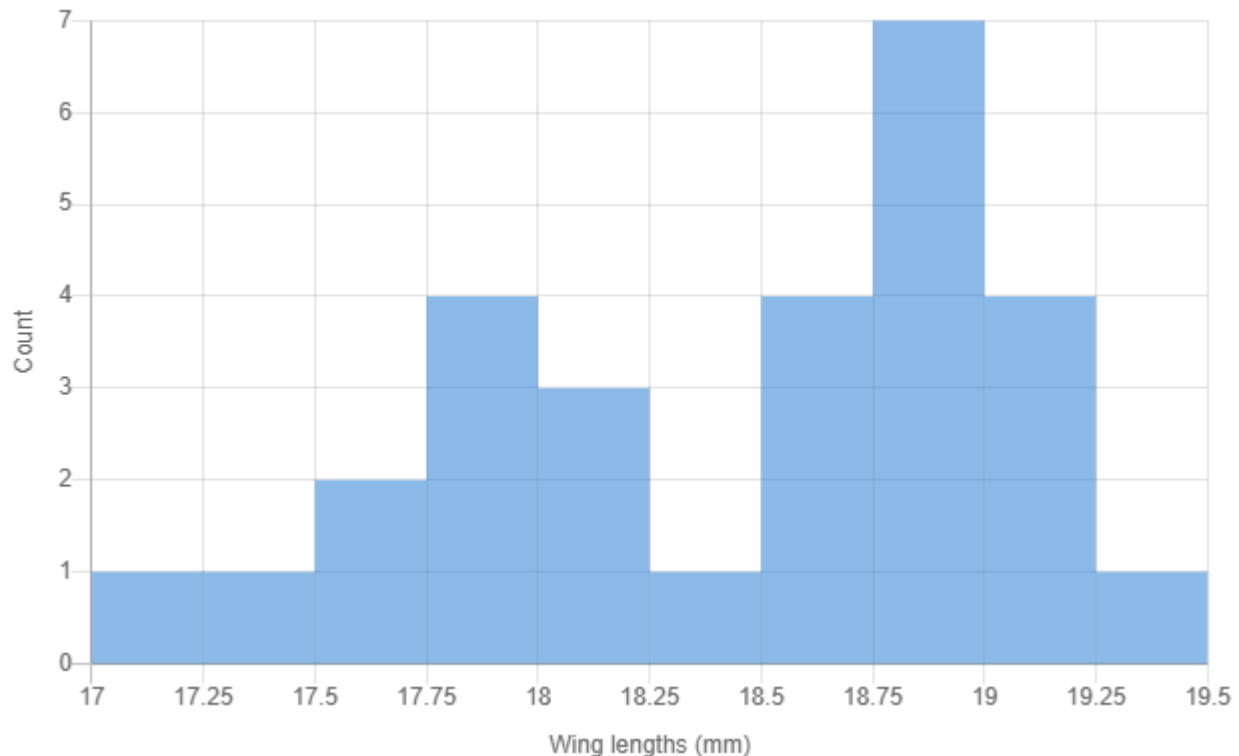
4. A football team is keeping careful track of how many yards they gain and how many points they score in each game they play. Based on their data, they have constructed the regression line below to predict points scored based on yards gained in a game:

$$\hat{y} = -9.58 + 0.106x$$

In one of their games, the team gained 378 yards and scored 24 points. Which of the following statements is true about the relationship between the point corresponding to this game and the regression line?

- (A) The residual for this point is large enough for us to call this point an outlier.
 - (B) Based on this point and its residual, the regression line does not effectively represent the trends in this dataset.
 - (C) The predicted value of the response variable is lower than the value we actually observed in this case.
 - (D) This point must have been added to the dataset after the regression line was constructed.
 - (E) **The regression line overestimates the value of the response variable for this observation.**
5. A faculty group wants to determine whether an administrator's job rating X (a numerical score conferred by their supervisor) is a useful linear predictor of the amount Y of their annual raise, in dollars. Thus they have constructed the following least-squares regression line: $\hat{y} = 14,000 + 2,000x$. Is there a relationship between the raises administrators at State University receive and their job performance? Choose the correct interpretation of the slope of this line from the options below.
- (A) For an administrator with a rating of 1.0, we estimate his/her raise to be \$2,000.
 - (B) For a 1-point increase in an administrator's rating, we estimate the administrator's raise to decrease by an average of \$2,000.
 - (C) **For a 1-point increase in an administrator's rating, we estimate the administrator's raise to increase by an average of \$2,000.**
 - (D) For a \$1 increase in an administrator's raise, we estimate the administrator's rating to increase by an average of 2,000 points.
 - (E) For a \$1 increase in an administrator's raise, we estimate the administrator's rating to decrease by an average of 2,000 points.

6. Which of the following changes **can** lead to a change in the correlation coefficient between two numerical variables X (explanatory) and Y (response)?
- (A) Adding 10 to every observation of X
 - (B) Multiplying every observation of Y by 5
 - (C) Expressing observations of X with different units (e.g., feet or meters)
 - (D) Deciding to consider Y as the explanatory variable and X as the response variable
 - (E) **Removing one observation (x_i, y_i) from the data set**
7. Ecologists in Greenland investigated the effects of rising summer temperatures on the wing lengths of a local butterfly species by recording summer temperatures each year as well as the wing lengths of butterflies born in those summers. The following histogram summarizes the data on butterfly wing lengths (in millimeters) collected by the ecologists. Which of the descriptions below accurately represents the shape of the histogram?



- (A) Unimodal, left skew
- (B) **Bimodal, left skew**
- (C) Unimodal, right skew
- (D) Bimodal, right skew
- (E) Uniform, symmetric

8. You are interested in gauging if there is a relationship between a person's eye color and whether or not they need glasses in a certain population of 165 individuals. Which of these is the best way to do this, using the contingency table below?

	No Glasses	Glasses	Total
Brown	45	30	75
Blue	20	35	55
Green	15	20	35
Total	80	85	165

- (A) Compute the correlation coefficient between the “No Glasses” and “Glasses” columns, and determine whether it is positive or negative.
- (B) Find the five-number summary for the “No Glasses” column, do the same for the “Glasses” column, and compare the results.
- (C) **Calculate the percentages of glasses-wearers among each eye color, and compare these values to the percentage of glasses-wearers in the total population.**
- (D) Graph all the numbers shown in the table as a histogram, and look for any skew.
- (E) None of the above; contingency tables are not appropriate for determining relationships between categorical variables.
9. Consider the following numerical data set:

$$\{2, 4, 5, 6, 7, 8, 9, 10, 15, 20\}$$

If we represented these data in a box plot, to which value would the **upper whisker** extend?

- (A) 10
- (B) **15**
- (C) 17
- (D) 17.5
- (E) 20
10. For a given set of observations of an explanatory variable X and a response variable Y , you have constructed the least squares regression line $\hat{y} = 0.85 - 0.24x$ and computed $R^2 = 0.81$. Which statement below **must be false** based on these figures?
- (A) The regression line approximates the linear relationship between the variables X and Y .
- (B) **The correlation coefficient between the values of X and Y used to fit this regression line is 0.9.**
- (C) When the value of X increases by one unit, the predicted value of Y decreases by 0.24 units.
- (D) When X equals 0, the predicted value of Y is 0.85.
- (E) 81% of the sample variance of observed Y values can be explained by this regression line.

11. The waiting times (in minutes) of nine customers at a coffee shop are listed below:

$\{4, 5, 5, 6, 6, 7, 7, 8, 9\}$

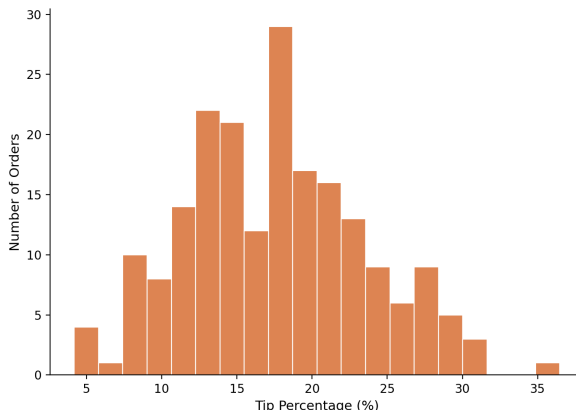
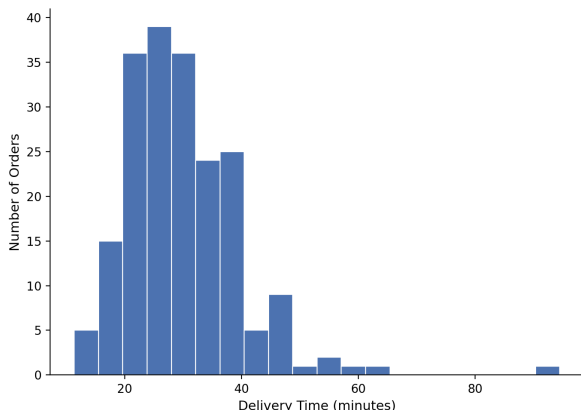
Suppose a data-entry error changes the final value from 9 minutes to 99 minutes. Which statement correctly describes what happens to the mean and median?

- (A) The mean and median both increase by 10 minutes.
 - (B) **The mean increases from about 6.33 minutes to about 16.33 minutes, while the median remains 6 minutes.**
 - (C) The mean remains about 6.33 minutes, while the median increases to about 16.33 minutes.
 - (D) The mean and median both become about 16.33 minutes.
 - (E) Neither the mean nor the median changes, because only one observation was changed.
12. A large city has a spreadsheet of data on 80 public parks, with one row for each park. For each park, they have recorded its acreage, number of playgrounds, and condition (poor, fair, good, or excellent). A researcher asks whether park acreage can be used to predict the number of playgrounds. Which of the statements below is correct?
- (A) **The cases are the 80 parks; acreage is the numerical explanatory variable, and number of playgrounds is the numerical response variable.**
 - (B) The cases are the playgrounds; number of playgrounds is the explanatory variable, and acreage is the response variable.
 - (C) The cases are the 80 parks; condition rating is the explanatory variable, and acreage is the response variable.
 - (D) Acreage is a categorical-ordinal variable because parks can be ranked from smallest to largest.
 - (E) Number of playgrounds is a numerical-continuous variable because it is recorded using numbers.
13. City officials have noticed that neighborhoods with a higher number of ice cream shops (X) also tend to have higher crime rates (Y); in particular, they have computed a correlation coefficient of 0.65 between the two variables. On this basis, one city councillor has proposed shutting down ice cream shops to reduce crime. Choose the most statistically sound objection to this proposal from the options below.
- (A) A correlation coefficient of 0.65 is too weak to justify any policy discussion whatsoever, regardless of the underlying explanation.
 - (B) Only a negative correlation coefficient between the two variables would be a cause for concern.
 - (C) **The relationship is likely driven by one or more confounding variables (e.g., population density) that independently increase both the number of ice cream shops and crime rates.**
 - (D) Number of ice cream shops is a discrete numerical variable, so correlation should never have been calculated with it in the first place.
 - (E) The correlation must be a coincidence, since ice cream and crime have no logical connection of any kind.

2 Free Response

1. A delivery platform studies the relationship between delivery time in minutes (x) and the customer's tip percentage (y) across 200 orders. A histogram of delivery times and tip percentages are given below. Using notation as in class, the summary statistics are

$$\bar{x} = 30, \quad s_x^2 = 100, \quad \bar{y} = 18\%, \quad s_y^2 = 36, \quad r = -0.4.$$



- (a) (10 points) The **delivery time distribution** is right-skewed (left-skewed / right-skewed / symmetric). The median (mean / median) is a better measure of its central tendency, while the interquartile range (standard deviation / interquartile range) is a better measure of its spread because median & IQR are robust; the distribution is right skewed with a potential outlier (justify using the shape of the distribution). Furthermore, the mean is greater than (greater than / less than / approximately equal to) the median.

The **tip percentage distribution** is symmetric (left-skewed / right-skewed / symmetric). The mean (mean / median) is a better measure of its central tendency, while the standard deviation (standard deviation / interquartile range) is a better measure of its spread because the distribution is symmetric with no outliers (justify using the shape of the distribution). Furthermore, the mean is approximately equal to (greater than / less than / approximately equal to) the median.

- (b) (6 points) Find the least-squares regression line predicting tip percentage from delivery time, and interpret the slope in context.

Solution: Step 1: Find s_x and s_y from the given variances.

$$s_x = \sqrt{s_x^2} = \sqrt{100} = 10, \quad s_y = \sqrt{s_y^2} = \sqrt{36} = 6$$

Step 2: Compute the slope using $\hat{b}_1 = r \frac{s_y}{s_x}$.

$$\hat{b}_1 = r \frac{s_y}{s_x} = -0.4 \cdot \frac{6}{10} = -0.24$$

Step 3: Compute the intercept using $\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$.

$$\hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x} = 18 - (-0.24)(30) = 25.2$$

So $\hat{y} = 25.2 - 0.24x$.

Slope interpretation: for each additional minute of delivery time, predicted tip percentage decreases by about 0.24 percentage points on average.

- (c) (3 points) Calculate R^2 and the percentage of the variation in tipping behavior is left **unexplained** by delivery time.

Solution: $R^2 = (-0.4)^2 = 0.16$. So delivery time explains about 16% of the variation in tip percentage, meaning 84% of the variation in tipping behavior is left unexplained by delivery time alone.

- (d) (4 points) One order took 90 minutes to arrive due to a snowstorm, and the customer tipped 3.5%. Classify this point as a residual outlier, high leverage point, or influential point. Justify your answer.

Solution: At $x = 90$: $\hat{y} = 25.2 - 0.24(90) = 3.6\%$, which matches the actual tip closely. **High leverage point.** Since the actual tip is close to what the line predicts (small residual), it does not pull the line noticeably and so it is *not* influential, and since it is not off-trend, the point is not a residual outlier either.

- (e) (4 points) A new order took 180 minutes to deliver. What does the model predict the tip percentage to be? What words of comfort would you give to the delivery partner who is worried that they may get very low tips as predicted by the linear model?

Solution: $\hat{y} = 25.2 - 0.24(180) = -18.0\%$. A negative tip percentage is impossible. A delivery time of 180 minutes is far outside the range of the data used to build the model, this is **extrapolation**. The model was never built to make predictions that far out; there is no reason for them to worry based on this number.

2. A STOR 155 professor wants to know whether bringing free donuts to 08:00 AM lecture *causes* an increase in attendance, or whether donut days just happen to coincide with days students would have shown up anyway.

- (a) (4 points) Design a randomized experiment to determine whether donuts have a causal effect on attendance. Be specific about how days are assigned to “donut” vs. “no donut,” and what is compared.

Solution: Each morning, by a coin flip, randomly assign lecture days to either “donut” or “no donut”. Bring donuts only on the days assigned to the donut group. At the end of the semester, compare the attendance rate on donut days to the attendance rate on no-donut days. Random assignment ensures that other factors affecting attendance (exam proximity, weather, time in the semester) are balanced on average across both groups, so any difference in attendance can be attributed to the donuts themselves.

- (b) (4 points) The professor happened to bring donuts on some days and not others, based on whatever mood struck that morning (not randomized). Each cell below counts the **number of lecture days** falling into that category, where “high attendance” means at least 80% of enrolled students were present that day:

	Attendance $\geq 80\%$	Attendance $< 80\%$	Total
Donuts brought	40	10	50
No donuts	30	20	50
Total	70	30	100

Calculate the percentage of days with attendance $\geq 80\%$, separately for donut days and non-donut days. Does bringing donuts appear to be associated with higher attendance? Explain

why this data alone cannot establish that donuts *caused* the higher attendance, and how the design from part (a) addresses that limitation.

Solution: Donut days: $40/50 = 80\%$ had high attendance. Non-donut days: $30/50 = 60\%$ had high attendance. Since $80\% > 60\%$, bringing donuts **does** appear to be associated with higher attendance. However, this data is observational and so no causal claims can be made. Randomizing the choice of donut vs no donut days removes confounding.